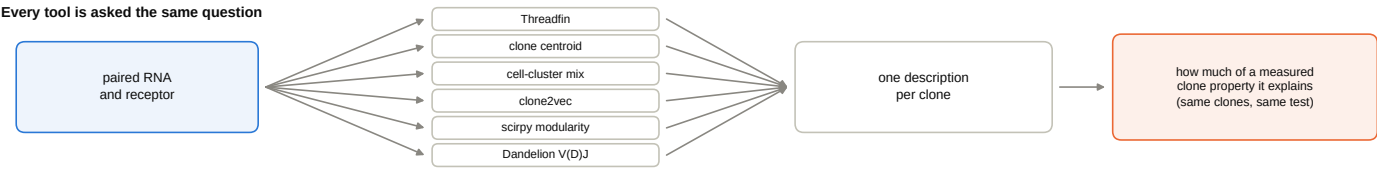


Supplementary Figure 4 | What other tools can and cannot do

Every published method that integrates single-cell BCR with gene expression, asked the same question on the same clones.

a

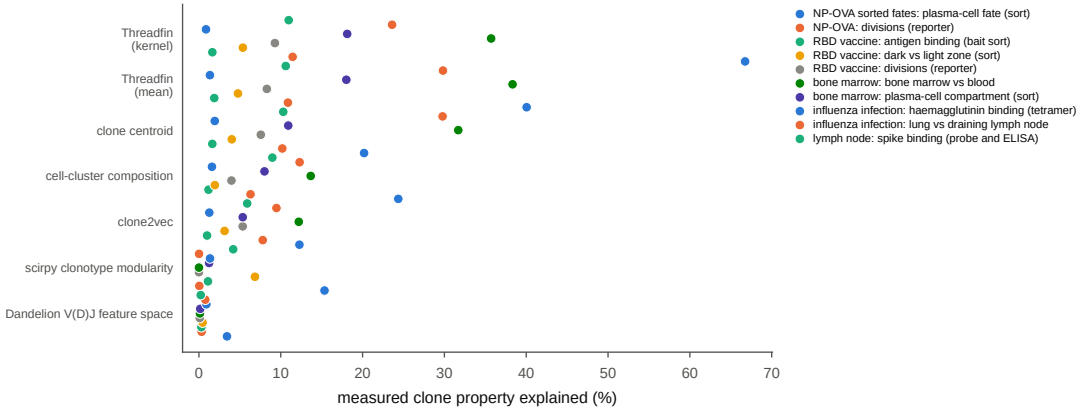


b

Threadfin	yes	yes	yes	yes	yes	yes
scripy (Bioinformatics 2020)	yes	no	partly	no	no	no
Dandelion (Nat Biotechnol 2024)	yes	no	no	no	no	no
CoNGA (Nat Biotechnol 2022)	partly	no	partly	no	no	no
Benisse (Nat Mach Intell 2022)	yes	no	partly	no	no	no
sciCSR (Nat Methods 2023)	no	no	no	no	partly	no
BIGCN (Small Methods 2026)	yes	no	partly	no	no	no
	clones from hypermutated receptors	is state inherited within clones?	a description of each clone	are there groups, or a continuum?	do clones keep their state?	a test with clones as replicates

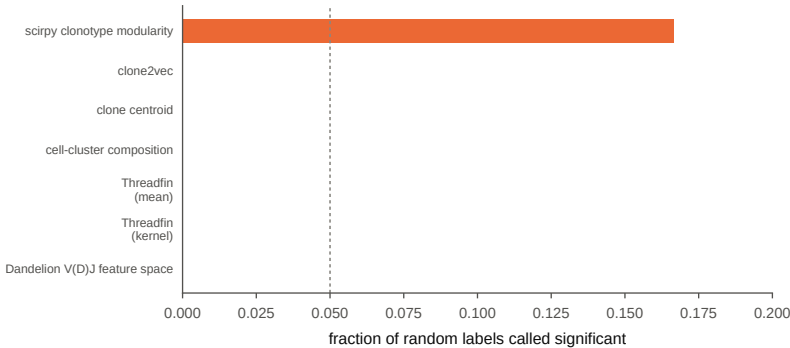
Only tools that integrate single-cell BCR with gene expression and were published in journals with an impact factor of at least five are shown. "Partly" means the tool provides something related but not the quantity itself.

c



Each point is one dataset and one experimentally measured clone property. All methods are scored on the same clones (at least three cells each) with the same statistic, and every method was also run on a label shuffled among clones.

d



A method that reports structure for a label flipped at random is not usable for discovery. Dashed line: the nominal 5%.